∴ Here, the dimension of Nul A = 3.

Observe how each basis vector corresponds to a free variable in the equation $Ax = 0$.

So, to find the dimension of Nul A, simply identify and count the number of free variables in $Ax = 0$. ☐

Definition: The **rank** of a matrix A, denoted by rank A, is the dimension of the column space of A.

Since the pivot columns of A form a basis for Col A, the rank of A is just the number of pivot columns in A.

Example 3: Determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 1 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 1 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \xrightarrow[2 \times \text{Row } 1]{\text{Row } 2 = \text{Row } 2 -} \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \xrightarrow[-3 \times \text{Row } 1]{\text{Row } 3 = \text{Row } 3}$$

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \xrightarrow[2 \times \text{Row } 2]{\text{Row } 3 = \text{Row } 3 -} \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \xrightarrow[3 \times \text{Row } 2]{\text{Row } 4 = \text{Row } 4 -}$$

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & -10 & 15 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row } 3 = \text{Row } 3} \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & -10 & 15 \end{bmatrix} \xrightarrow[\times 1/5]{\text{Row } 4 = \text{Row } 4}$$

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & -2 & 3 \end{bmatrix} \xrightarrow[+\text{Row } 3]{\text{Row } 4 = \text{Row } 4} \begin{bmatrix} \boxed{2} & 5 & -3 & -4 & 8 \\ 0 & \boxed{-3} & 2 & 5 & -7 \\ 0 & 0 & 0 & \boxed{2} & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Its easy to see that Row 1 Column 1, Row 2 Column 2 and Row 3 Column 4 are pivot positions. ∴ Columns 1, 2 and 4 are pivot cols.